Kanishak Mittal

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Education

Indian Institute of Information Technology Raichur — Raichur, Karnataka
B.Tech in Computer Science and Engineering; ***CGPA: 9.47*** — *Aug. 2024 – May 2028*
- **Relevant Coursework:** Design and Analysis of Algorithms, Computer Architecture and Organization, Probability, Data Warehousing, Database Management Systems, Operating Systems, Compiler Design

Delhi Public School, Eldeco — Lucknow, UP
Class XII (12th); ***Percentage: 95.2%*** — *2024*

Delhi Public School, Eldeco — Lucknow, UP
Class X (10th); ***Percentage: 95.8%*** — *2022*

Experience & Leadership

Summer Research Intern — Jun. 2026 – Jul. 2026
UbiNet Lab, Indian Institute of Technology Kharagpur — *Kharagpur, India*
- **Dynamic Observability Framework** (Sponsored by Ericsson) | *eBPF, Kubernetes, MicroVMs, C++, Python*
- Engineered a zero-instrumentation observability framework using eBPF (kprobes/tracepoints) to trace distributed transactions and kernel-level latency at near-zero steady-state overhead
- Architected container lifecycle tracking across Kubernetes, Docker, and MicroVMs, using dynamic metadata discovery (CRI runtime, procfs) to resolve ephemeral namespaces, host PIDs, and IPs
- Restructured eBPF map memory layouts to eliminate locked-memory overhead and prevent OOM kills under load

Science and Technology Secretary — Aug. 2025 – Present
Indian Institute of Information Technology Raichur — *Raichur, Karnataka*
- Spearhead the functioning and coordination of all science and technology clubs across the institute
- Organized and executed competitive programming competitions to foster a coding culture among students
- Lead the organization of a state-level event under the AI Impact Summit 2026, managing logistics and execution

Core Member, Cybersecurity Club — Aug. 2024 – Present
Indian Institute of Information Technology Raichur — *Raichur, Karnataka*
- Actively participate in club activities and contribute to the planning of cybersecurity-focused events and workshops

Projects & Research

Muadib: Which Exposure Matters? | *Python, PyTorch, CUDA, NumPy* — Oct. 2026 – Present
- Quantifying how predicate-, pair-, and rule-level exposure affects compositional generalization in diffusion models
- Built a synthetic scene renderer, Monte Carlo exposure estimator, LP-based exposure design, and a 127+ test suite
- Developed a CUDA diffusion pilot with a custom UNet, predicate-slot cross-attention, and classifier-free guidance

Goodix 27c6:581a Linux Fingerprint Driver | *C/C++, Linux Kernel, libfprint, Python* — Oct. 2026 – Nov. 2026
- Reverse-engineered a proprietary USB fingerprint sensor into a `libfprint/fprintd` biometric driver for Linux
- Reimplemented the OEM TLS 1.2 handshake over USB bulk endpoints and decoded packed 12-bit frames into native 64×80 images with a separable high-pass filter
- Replaced NBIS/Bozorth3 minutiae matching with a SIGFM SIFT matcher; hardened the stack with SELinux policy fixes

GigTrust | *Flutter, Rust, JavaScript, Node.js, MongoDB* — Aug. 2025 – Dec. 2025
- Developed a blockchain gig-work platform with Rust smart contracts that freeze funds until mutual agreement
- Built a cross-platform Flutter application with a Node.js and MongoDB backend

Laxmi | *Python, AWS* — May 2025 – Jan. 2026
- Built an algorithmic trading bot that executes real-time trades via mTrade REST APIs with custom strategies
- Deployed and currently maintain the bot infrastructure on Amazon Web Services (AWS)

Technical Skills

Languages: C/C++, Python, Rust, JavaScript, Dart, SQL
Frameworks & Tools: PyTorch, Flutter, Node.js, Git, AWS, Docker, Kubernetes, eBPF
Databases: MongoDB, PostgreSQL